

Introduction to Operating Systems

Sam Ogden

DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Class Overview

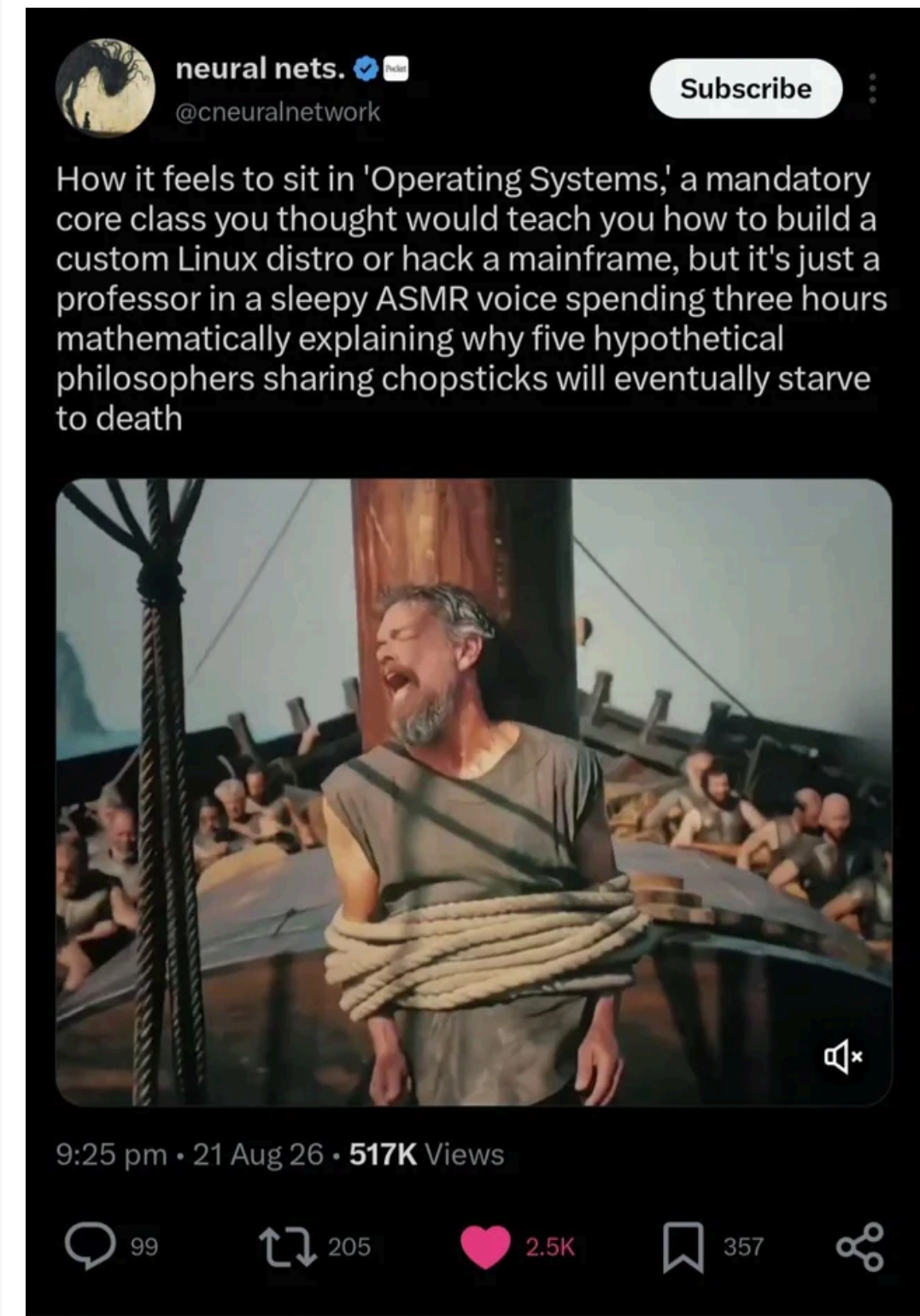
What are Operating Systems

Systems is how we think about how multiple programs work together

This course focuses on those that run on a single machine, but is applicable in many different places!

Course Learning Outcomes

- By the end of this class I want you to know how to reason about trade-offs made in system design
- You should be able to think about the pros and cons made in different kinds of systems:
 - Process Management
 - Memory Management
 - Concurrency and Parallelism
 - I/O and Hardware Interactions
 - Programming Language Design
 - Security for OSes
- As well as being conversant in the C programming language



Teaching Staff

Me (Dr. Sam Ogden)

- BS (Math&EE), Univ. of Vermont
- MS (CS), Univ. of Vermont
- PhD (Comp. Sci.), WPI

- Product Engineer @ IBM/GF (5 years)
- Grad Student @ WPI (5 years)
- Professor @ CSUMB (since 2022)

Course TAs



Ryan Hopper (TA Fall 2026)



Edwin Kofler (TA Fall 2026)

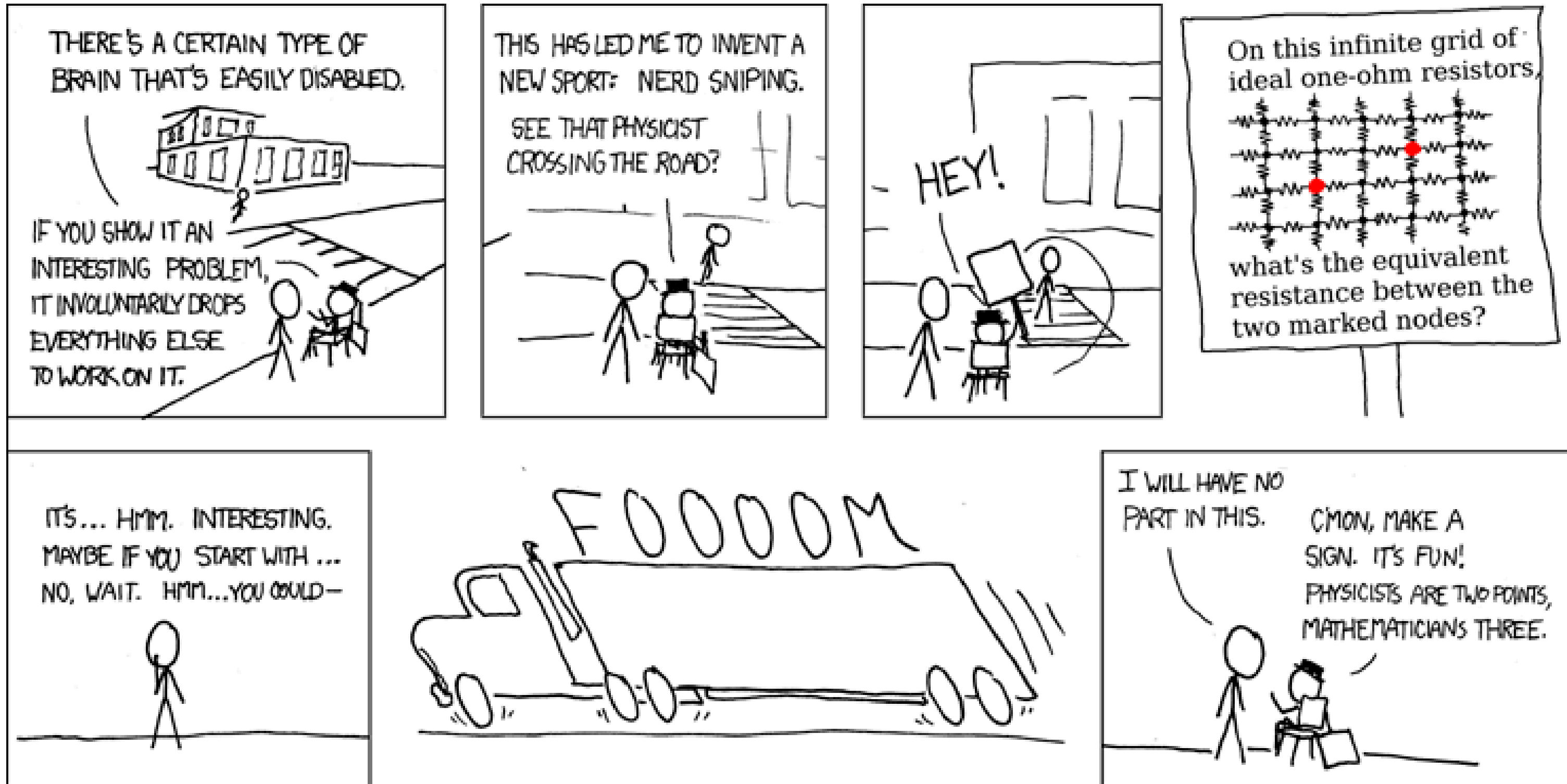
Asking Questions

I love getting asked questions

During class please raise your hand and I'll get to you as soon as I can

If you're confused then chances are that you're not the only one

Questions: Nerd Sniping



Cat Tax



Course Information

Textbook: OSTEP

Course Schedule (might change a bit!)

Class Structure

The goal is for most days to have a lecture, a break and a lab:

- lecture (50 minutes)
- break (10 minutes)
- lecture (35 minutes)
- lab (15 minutes)

(but if you've ever had a class with me you know this is only a goal)

Course Policies

[Link to the Syllabus](#)

Grading

Assignment Kind	Value
Homework	40%
Exams	40%
Participation	20%

You must get a 60% overall, and 40% in each subset of these in order to pass

Attendance

Attendance is expected and enforced through quizzes

Use of LLMs (1)

My personal thoughts on LLMs can be summarized as:

- LLMs are very useful tools for accelerating our work.
- LLMs are fantastic at making “interactive data”
- LLMs can be incredibly stupid. (And see the first bullet...)

One of the things are is most valuable about us is **our ability to problem solve.**

But we have to **learn how** and **keep practicing** it

Use of LLMs (2)

How I expect you to use LLMs in this class:

- Debug computer problems
- Better understand the material
- Quick iteration
- Checking your approach and assumptions

How I expect you to *not* use LLMs in this class:

- Abdicating decisions and/or thinking
- Taking the class for you
- Copy-and-paste

The point of you being in this class is to understand the topic, right?

What gets you a job is the ability to think and put pieces together.

If I ask you to explain your code or analysis, you should be able to do so.

Assignment Details

Programming Assignments

OS is not a programming class – it's a concept class

PAs are meant to give context to what we are learning

- Assignments are in the github repository
- You should only change `student_code.c` and `student_code.h`
 - All other files submitted are discarded
- 6-7 assignments throughout the class
 - each assignment is 2 weeks long
- They are in the C programming language
 - C is meant to be very simple – often confusingly so
 - Very similar to Java and C++
 - Drawing out memory contents can often help a lot!

Programming Assignment: Critical Info

- The most important thing to do is to **make sure your code compiles**
 - The starter code will compile
 - If your code doesn't compile then you can't get any points at all
- The autograder runs **2x a day** (at noon and midnight)
 - It will determine your grade, or tell you if something went wrong

Exams (30%)

- 4 equally weighted exams
- closed book, but 1 handwritten page of notes
- Material builds, so effectively cumulative
- Practice Quizzes will be provided beforehand

Participation (20%)

- Class Attendance (10%):
 - many classes start with attendance quizzes
 - reading is linked on weekly schedule
 - More likely to have quizzes when class looks empty
- Weekly Study Notes (5%)
 - Intended to help you write yourself a study guide for exams
 - Reflect on the topics covered this week
 - what you learned this week
 - what you struggled with
 - Graded largely on completion
 - The guidelines are meant as an algorithm – you can use whatever format you find most helpful to study from
- Lab activities (5%):
 - due Sunday night after in-class introduction

Weekly Study Notes: help your future self

These study notes are for you to go back and use as a basis for studying

How to approach these notes:

1. identify the topics that were covered
2. explain ideas in your own words
3. identifying least-understood topics
4. noting “aha” moments
5. explaining the nature of the confusion
6. asking questions
7. wondering about ideas, connections, applications
8. deeper reflection

Homework

How to get help

TA Office Hours!



How to get help: what's okay

- Asking your instructor!
 - Seriously, come by office hours
- Asking your TAs!
 - They've taken the class and are nice!
- Asking LLMs for clarification or guidance!
 - “What does limited direct execution mean?”
 - “Can you make me some practice problems for process scheduling?”

But don't trust LLMs entirely!

How to get help: what's not okay

- Getting `[fill in the blank]` to write code for you
 - Friends
 - Enemies (for numerous reasons this is a bad idea)
 - LLMs
- The goal is to understand the problems and learn how to approach them!
 - Any code you turn in you are responsible for!

Dealing with Challenges in Class

Take a breath

Mental Health

- Mental health is super important
- If you're feeling stressed or anxious about an assignment then take a break on it for a bit!
 - (This is the real reason I want you to look at assignments early — so you have time to walk away for a bit)
- If you're feeling stuck then ask me or the TAs!

How to solve hard problems

Give yourself time to solve problems

College has a lot going on, but starting early gives you time to step back

Sometimes we run into impossible problems and need to start over

Things to do

- Go for a walk without devices
- Go to the gym
- Stare at a wall
 - (boredom is your friend)
- Read a book for fun

Things to not do

- Screens.
 - (I really could stop there)
- Use your brain to focus heavily

Give yourself time and space to let your mind wander and make random connections

A quick note back to LLMs

This is one of the big traps with LLMs – they force you into *doing* rather than *thinking*.

- I've seen this myself and know a lot of pros are running into it, too

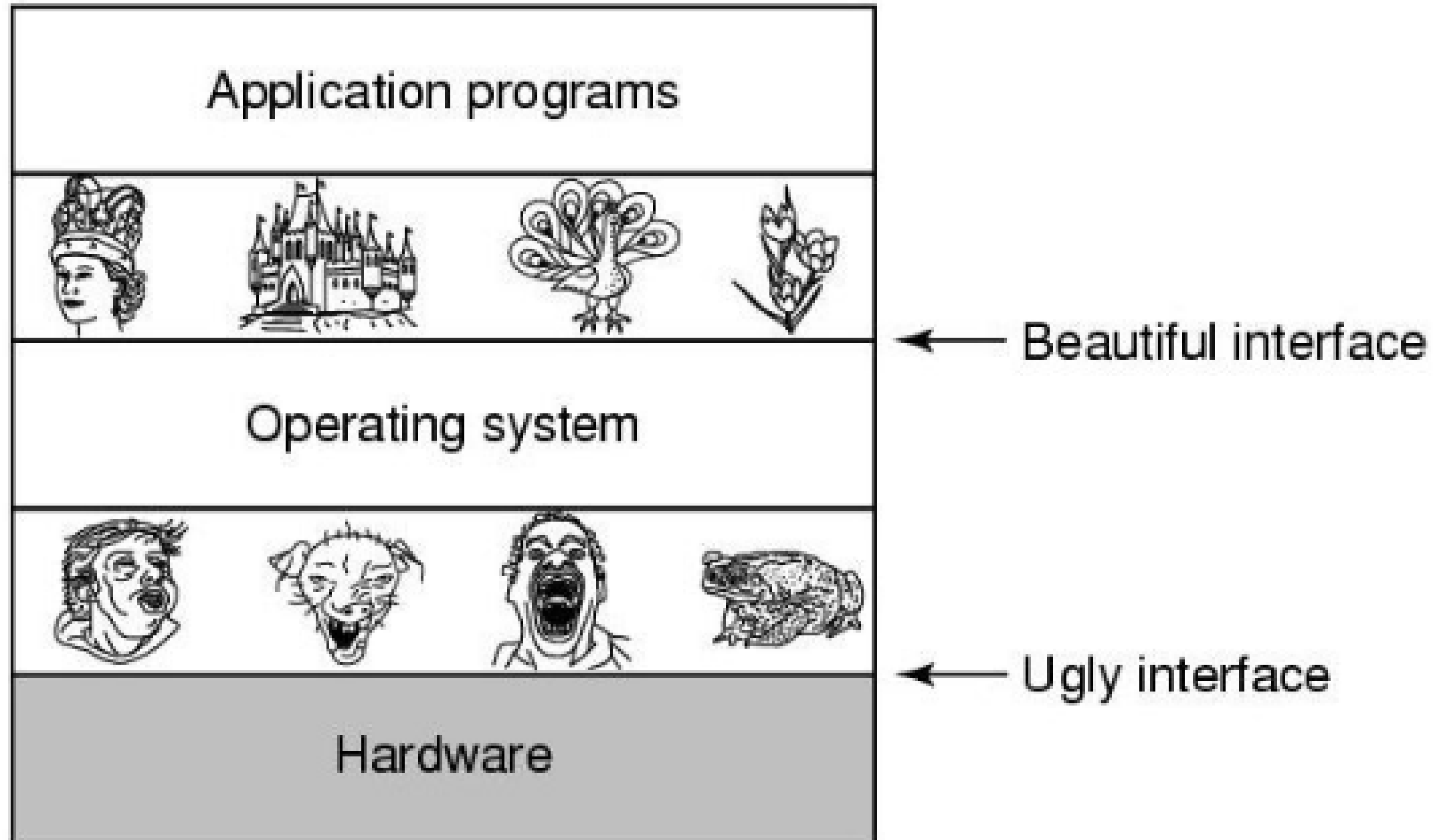
This causes burnout and poor learning

Some things to remember

- Focus on growth and mastery
- Start homework early
- Drop by my office to chat
- Please call me just Sam (or Dr./Prof. Ogden)

Operating Systems

What's an Operating System?



Tanenbaum

OS is a “systems” course

- In a CS systems course, we think about:
- physical resources shared by multiple people or applications
- What are the resources of a computer system?
 - CPU
 - Disk
 - Memory (RAM)
 - I/O



OS is a “systems” course

- How is this different from other programming classes?
 - We need to consider **how programs interact**
 - We our porgram is **no longer running in isolation**
 - In fact, we know it isn't!
 - We have **concrete limits** to resources!

The Core Ideas of OS

Problem Statement

Core Question:

How to provide resources to users in an easy-to-use, reliable, fair, efficient, and secure way?

- Operating Systems is an amazing subject because this problem is everywhere:
 - restaurants
 - gyms
 - colleges
 - roads
 - ...

Solution Approach for Systems

The approach for systems problems:

1. Begin with the minimal solution that solves the problem
2. Identify bottlenecks and add complexity only where necessary
3. Repeat step 2 until requirements are satisfied

The Core Idea of Systems (and OS)

Operating systems are designed to work and to be simple.

The most important thing to remember is we want to find the simplest solution that works well enough

An [imperfect] Example: cake

- Resource: A cake
 - For now, let's assume it's a round cake
- Users: 7 users total
- Challenge:
 - How do we divide up the cake?

Core ideas of OS

- Our discussions often have two parts
 - **Mechanism:** What low-level support is needed to make this happen
 - e.g. a knife for the cake example
 - **Policy:** How do we use the mechanism to achieve particular goals?
 - e.g. ensure everybody has an equal sized piece of cake

Example: Processor as a shared resource

Example Guiding questions

- What are the virtual resources or services offered to users? Are they easy to use?
- How to ensure fair sharing of resources between users?
- How to protect users from each other, and protect the system from users?
- What workloads do we use to measure performance?
- What metrics do we use to measure performance?
- How efficiently are resources managed?

Recap

- Welcome to OS!
- Three main chunks of assignments
 - Programming Assignments are graded 2x a day
- Take time with this class
 - If you leave everything to Sunday night you're gonna have a bad time
- When we approach problems, let's think about how we can do thing *simply*
 - And *good enough* is enough